Session 3 — What "AI" means, types of AI, and how machines learn

Artificial Intelligence · 4th year · UIR · Pr. Hakim Hafidi

Study sheet — exam material

1. The thread of this session

Two sessions of history have shown where AI comes from. This session closes the foundations before the course turns to language models: what the word actually means, what kinds of AI exist, what *learning* means for a machine, and how we tell whether a model has learned anything at all.

The last part is the one that matters most for the rest of the semester. Everything in Labs 3 and 4 — and every number in your project report — depends on one idea: **a model is only worth what it is worth on data it has never seen.**

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2. Defining AI: four answers and the one we use

No definition of artificial intelligence is universally accepted, because no definition of intelligence is. What exists is a set of useful framings.

Four historical definitions. Minsky (1968): "the science of making machines do things that would require intelligence if done by men". Kurzweil (1990): "the art of creating machines that perform functions that require intelligence when performed by people". Russell and Norvig (2020): "the study of agents that receive percepts from the environment and perform actions". And Larry Tesler's wry theorem: AI is "whatever hasn't been done yet".

The Russell & Norvig quadrants. A standard way to organise the field uses two axes: is the standard *human* or *rational*, and are we modelling *thought* or *behaviour*?

	Human standard	Rational standard
Thought	Thinking humanly — cognitive science, modelling the mind	Thinking rationally — logic, Aristotle, symbolic AI
Behaviour	Acting humanly — the Turing test	**Acting rationally — the rational agent (this course)**

The rational agent, which is our definition. An agent perceives its environment through sensors, decides, and acts on it through actuators, choosing the action expected to best achieve its goal given what it knows. Note *rational*, not *perfect*: rationality is bounded by available information and computation.

A thermostat is an agent under this definition. So is a chess program, a self-driving car, and an LLM assistant with tools. The definition says nothing about neural networks — and it comes back, almost word for word, in Session 8 when "AI agent" becomes a concrete engineering term.

The nested circles. AI ⊃ machine learning ⊃ deep learning ⊃ generative AI. There is AI without learning (search, planning, expert systems), learning without neural networks (decision trees, regression), and neural networks that generate nothing (classifiers).

The Chinese Room (Searle, 1980). A person who speaks no Chinese sits in a room with a rule book. Chinese characters come in; the person matches them against the book and passes characters out. From outside, the room answers perfectly. Searle's claim: the room passes the Turing test, yet nobody in it understands Chinese — so **syntax is not semantics**, and symbol manipulation alone cannot constitute understanding.

Standard replies: perhaps the *system as a whole* understands, even if the person does not; or perhaps understanding simply *is* sufficiently good symbol manipulation, and our intuition is wrong. Forty-five years later this is the central debate about large language models, and it is not settled.

Moravec's paradox (1988). What is hard for us is easy for machines, and the reverse. Chess at champion level, multiplying twenty-digit numbers, searching a billion documents: easy for machines. Walking on uneven ground, folding laundry, recognising a friend from behind, understanding a joke: hard. The explanation is evolutionary — abstract reasoning is recent and shallow in us, while perception and movement have been optimised over a billion years.

The AI effect. As soon as a capability works reliably, it stops being called AI and becomes "just software". Chess in 1997, spam filters in the 2000s, speech recognition around 2010, translation around 2016, chatbots in 2022. The field keeps its name for the frontier.

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3. Types of AI

By capability.

Narrow AI (ANI) — excellent at one task or one family of tasks, with no transfer. AlphaGo cannot play chess; a self-driving car cannot pilot a

boat. **Every system that exists today is narrow**, including large language models: they handle thousands of tasks, but all of them are text-shaped and inside their training distribution. Some call this "jagged intelligence": superhuman peaks beside sub-human valleys.

General AI (AGI) — a hypothetical system able to learn any intellectual task a human can. There is no agreed test. Predictions have ranged from "within five years" to "never", continuously, since 1956.

Superintelligence — hypothetical, exceeding the best humans in every domain; the object of long-term safety research.

By approach. *Symbolic* (knowledge and rules, explainable, brittle), *connectionist* (weights learned from data, robust to messy input, opaque), and *hybrid or neuro-symbolic* (a neural model calling a calculator, a planner, a database). Sessions 7 and 8 are hybrid by design.

By function, the classic textbook taxonomy. Reactive machines (no memory — Deep Blue, a spam filter); limited memory (use the recent past — self-driving cars, chatbots: nearly all current AI); theory of mind (would model other agents' beliefs — a research topic); self-aware (science fiction). Note that nothing guarantees level 3 follows level 2.

By task — the three colours of this course.

	Period	What it does	Examples
Predictive AI	2010s–	Predicts a label or a number from data	Fraud scoring, forecasting, medical imaging
Generative AI	2022–	Produces text, images, code, audio	Chatbots, copilots, image tools
Agentic AI	2025–	Pursues a goal by taking actions with tools	Coding agents, research agents

Discriminative versus generative. A discriminative model draws a boundary: given an input, decide the category or value. A generative model learns the whole distribution and samples new content from it. Same underlying technology; different task.

4. What learning means for a machine

The four ingredients. Data (examples, features, labels), a model (a function with adjustable parameters), an error measure, and a procedure for reducing it.

Data vocabulary. An *example* is one row — one house, one patient, one email. *Features* are its measurable properties — size, number of rooms, district, age. The *label* is what we want to predict — the price. Images, text and sound are data too: pixel values, sequences of tokens, amplitudes.

The model. The simplest useful model is a weighted sum: price ≈ w_1 × size + w_2 × rooms + w_3 × age + b. The w's and b are the **parameters**: the numbers the model is free to change. This is the perceptron of Session 2 without the threshold. A deep network is the same idea repeated in layers — from four parameters to a trillion, the learning principle is identical.

The learning loop. Predict with the current parameters. Measure the error against the truth. Nudge each parameter in the direction that reduces the error. Repeat over thousands of examples. Nobody tells the model which parameter to turn: the error itself points the way, and computing that direction for every parameter at once is what backpropagation does.

Gradient descent, without formulas. Picture the error as a landscape whose altitude is how wrong the model is and whose position is the setting of the parameters. You cannot see the whole landscape, but you can feel the slope under your feet. Take a small step downhill, feel again, repeat. Real landscapes have billions of dimensions and many valleys; the process settles in a good-enough one, not necessarily the best.

Training versus inference. *Training* adjusts the parameters on data; it happens once (or rarely), takes hours to months, needs the labels, and produces a file of numbers — the weights. *Inference* applies the frozen model to a new input; it happens millions of times a day, is fast and cheap, and **the parameters do not change**. Chatting with an LLM is inference: the claim "the model learns from our conversation" is almost always false.

Parameters versus hyperparameters. Parameters are learned by the machine. Hyperparameters are chosen by you: number of layers, size of the model, learning rate, batch size, number of epochs, which architecture, which data. Analogy: the oven setting is a hyperparameter; the cook's acquired touch is the parameters.

Vocabulary you will hear. *Loss* is the error. *Gradient* is its slope. An *epoch* is one pass over the training data. A *batch* is a group of examples processed together. The *learning rate* is the step size. *Convergence* is when the error stops improving.

5. The four learning paradigms

Supervised learning. Every example comes with the right answer: this email is spam, this image is a cat, this house sold for 2.1 million. The model learns the mapping from input to answer. It is the most common paradigm in industry and the most expensive, because someone has to

label the data. Two sub-types: *classification* when the label is a category, *regression* when it is a number. Analogy: a workbook with the answers at the back.

Unsupervised learning. Inputs only, no labels. The model finds structure: *clustering* groups similar items (customer segments, similar documents, anomalies that fit no group); *dimensionality reduction* compresses data to its main directions. Cheap, because nothing must be labelled, but hard to evaluate — what is a "good" cluster? Analogy: sorting a box of unlabelled photographs into piles that look alike.

Reinforcement learning. No labels and no fixed dataset. An agent takes actions in an environment and receives rewards — points, a win, a penalty. It learns a *policy*: what to do in which situation. The hard part is **credit assignment**: the reward arrives late, so which of a hundred moves deserves it? Used for games (AlphaGo, AlphaZero, Atari), robotics and control — and, crucially for this course, for the alignment step that turned GPT-3 into ChatGPT. Analogy: training a dog with treats at the right moment.

Self-supervised learning. The trick behind every LLM. Hide part of the input and ask the model to predict it: take a sentence, remove a word, predict the word. The label is the data itself, so labelling is free and the dataset is effectively infinite — every sentence ever written is a training example. Technically it is supervised learning; what changes is that no human produced the labels. Analogy: fill in the blank.

Paradigm	The data	What is learned	Typical use
Supervised	Inputs + correct answers	A mapping input → answer	Diagnosis, spam, pricing
Unsupervised	Inputs only	Structure: groups, directions	Segmentation, anomalies
Reinforcement	Actions + rewards over time	A policy: what to do when	Games, robots, RLHF
Self-supervised	Inputs with parts hidden	To predict the hidden part	Pretraining LLMs

They combine. A modern LLM is pretrained self-supervised, fine-tuned supervised, then aligned with reinforcement learning. That is Session 5 in one sentence.

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6. Does it really learn? Generalization and its failures

The only question that matters. Does the model work on examples it has never seen? A model that scores perfectly on its training data has told you nothing.

Holding data back. Shuffle all your labelled data and split it once:

- **Training set (~70%)** — the model adjusts its parameters on these examples only.
- **Validation set (~15%)** — used to choose hyperparameters and decide when to stop. Seen many times, never trained on.
- **Test set (~15%)** — touched once, at the very end, for the honest score. If you look at it while developing and tune against what you see, the number it gives you is a lie.

Overfitting. A model with enough capacity will memorise the training data, including its noise. The signature is unmistakable once you have seen it: training error keeps falling while error on held-out data starts rising. Remedies: more data, a simpler model, stopping training earlier, and regularization (penalising large weights). The opposite failure, **underfitting**, is a model too simple to capture the pattern — high error everywhere.

Bias. The model does not know what is fair; it reproduces the world in its data, faithfully. Three distinct mechanisms:

Historical bias — a hiring model trained on ten years of past hires learns the past pattern, including who was never hired. Amazon abandoned such a tool in 2018.

Representation bias — the Gender Shades study (Buolamwini and Gebru, 2018) found commercial face-classification error rates up to 35% for dark-skinned women against under 1% for light-skinned men, because the training sets were overwhelmingly light-skinned and male.

Measurement bias — Obermeyer et al. (2019) showed a widely used health system used *healthcare cost* as a proxy for *healthcare need*, which systematically under-served Black patients, on whom less had historically been spent.

Accuracy can lie. Consider a disease affecting 1% of patients and a model that is 99% accurate. It may be a model that always answers "healthy": it catches zero patients and is still right 99% of the time. With imbalanced classes, accuracy hides exactly what matters. Look at the confusion matrix — true positives, false positives, true negatives, false negatives — and at the two metrics derived from it: **recall** (of the true cases, how many did we catch?) and **precision** (of those flagged, how many were right?). Which error costs more depends entirely on the use: missing a cancer is not the same kind of mistake as a false spam flag.

Three further limits. *Correlation is not causation* — ice-cream sales predict drownings, because both rise in summer; a model finds the correlation and predicts happily. *Distribution shift* — trained on 2019 traffic, deployed in 2020 lockdown; trained on one hospital's scanners,

used in another; the world changes and the model does not know. *No goals of its own* — a model optimises literally the objective it was given; ask for engagement and you get outrage.

7. Key terms

Term	Meaning
Rational agent	Perceives through sensors, decides, acts through actuators to best achieve its goal given what it knows.
Chinese Room	Searle's 1980 thought experiment: rule-based symbol matching can pass the Turing test without understanding. Syntax ≠ semantics.
Moravec's paradox	Reasoning is easy for machines; perception and movement are hard — the reverse of humans.
AI effect	Once a capability works, it stops being called AI.
Narrow AI (ANI)	Excellent at one task or family of tasks, no transfer. Every system today.
AGI / superintelligence	Hypothetical general, then superhuman, capability. No agreed test.
Discriminative vs generative	Draw a boundary between classes vs learn the distribution and sample new content.
Example / feature / label	One row of data / its measurable properties / the target to predict.
Parameters	The numbers inside the model, learned from data.
Hyperparameters	Settings chosen by the engineer: layers, learning rate, batch size, epochs.
Training vs inference	Adjusting parameters on data vs applying the frozen model to new input.
Loss / gradient / epoch / batch	The error / its slope / one pass over the data / a group of examples.
Gradient descent	Repeatedly stepping downhill on the error landscape.
Supervised learning	Inputs with correct answers; classification (category) or regression (number).
Unsupervised learning	Inputs only; clustering and dimensionality reduction.
Reinforcement learning	Actions and rewards; learns a policy; credit assignment is the hard part.
Self-supervised learning	Hide part of the input and predict it; free labels; how LLMs are pretrained.
Train / validation / test	Data to fit / to choose settings / touched once for the honest score.
Overfitting / underfitting	Memorises noise / too simple to capture the pattern.
Regularization	Penalising complexity to reduce overfitting.
Confusion matrix	The table of true/false positives and negatives.
Precision / recall	Of those flagged, how many were right / of the true cases, how many were caught.
Distribution shift	The deployed data no longer resembles the training data.

8. Numbers and references to know

Typical split 70 / 15 / 15 · the 99%-accuracy trap on a 1% positive class · Gender Shades (2018): up to 35% error for dark-skinned women versus under 1% for light-skinned men · Amazon's abandoned hiring tool (2018) · Obermeyer et al. (2019) on cost as a proxy for need · Searle's Chinese Room (1980) · Moravec's paradox (1988) · Russell & Norvig's four quadrants.

9. Common misconceptions

"The model learns from our conversation." It does not. Chatting is inference; the weights are frozen. Anything that looks like memory was put back into the context by the application.

"99% accuracy means the model is good." Only if you know the class balance and the cost of each error type. On rare events, accuracy is close to meaningless.

"Overfitting means the model is too big." Size is one cause among several. Too little data, too many epochs and no regularization produce it just as reliably.

"Unsupervised learning needs no human work." No labelling, but choosing the representation, the number of clusters and the evaluation is all human judgement.

"AI is unbiased because it is mathematical." The model reproduces the statistics of its training data. The mathematics is neutral; the data is not.

10. Pause & recall — answers

I. (a) The Turing test belongs to *acting humanly*; this course takes *acting rationally*. (b) Matching symbols by rules is not understanding them; a standard reply is that the system as a whole understands, or that understanding just is very good symbol use. (c) Chess, spam filtering, face unlock, translation — all called AI until they shipped.

II. (a) Because it handles only text-shaped tasks inside its training distribution and cannot learn a new skill from a single lesson. (b) A neural model wired to symbolic tools — an LLM calling a calculator or a knowledge graph. (c) Spam filter: discriminative. Image generator: generative. Credit scoring: discriminative.

III. (a) Features: size, rooms, district, age. Label: price. Parameters: w_1, w_2, w_3 and b. (b) Training adjusts the parameters on many examples, once; inference applies the frozen model to one new example, many times. (c) Each parameter setting is a position, the error is the altitude; you feel the slope and step downhill, repeatedly.

IV. (a) Whether the examples come with the correct answer. (b) Because the reward is delayed and sparse, so credit must be assigned across many actions. (c) The label is the hidden word itself, taken from the text — no human is needed.

V. (a) To measure performance on data the model has never seen; training accuracy says nothing about generalization. (b) Perfect on training data, poor on new data; remedies include more data, a simpler model, early stopping and regularization. (c) If the positive class is 1%, always predicting "negative" is 99% accurate and catches nothing.

11. Sample exam questions

1. Define AI as this course does, and explain each part of the definition.
2. Give the four learning paradigms with one real example each, and explain why LLM pretraining counts as self-supervised.
3. A fraud model reaches 99.5% accuracy. Explain why this may be meaningless and name the metrics you would look at instead.
4. Explain overfitting with a sketch of the two error curves, and give three distinct remedies.
5. Distinguish historical, representation and measurement bias, with one documented example of each.